

Detecting Q-Dominance Ex-Ante for Streamed-Liability Convex Hedges:

An FX-Variance-Share Sensitivity Surface Methodology

Section 5 — E10 GSPS Application

Abrigo Analytics — Iteration `e10_gspv0.7`

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Abstract

We propose a methodology for detecting, ex-ante and from public-source data, whether a streamed-liability convex hedge addresses the dominant variance vector of a target cohort’s cost stream. The methodology is built around an exact three-way log-variance decomposition of the cost-stream identity $FX_t \cdot p \cdot Q_t$ into FX, Q, and covariance components on a common daily gapped grid; an FX-variance-share sensitivity surface evaluated on a log-spaced grid across an anchored Q-volume range; and a six-input collectively-exhaustive descriptive verdict ladder consuming surface, material-gap, simulator-anchored, Q-variance-dominance, interior-crossing, and data-visibility flags. The methodology emits one of SURFACE-PRODUCED, PARTIAL, or NON-RETIREMENT. Applied to the E10 representative Web3 data-analyst cohort over a 5-currency panel (COP, BRL, EUR, GBP, NGN; 150 monthly cells; 2023-11 to 2026-04), the empirical FX-variance share sits approximately 3.21 decades below the ex-ante-pinned break-even $s_{be} = 0.25$ at every grid point on the anchored $[28, 130]$ queries/month range; all five pre-committed sensitivity arms remain concordant; the verdict is NON-RETIREMENT, subtype `q.dominance`. The methodology contribution survives the specific verdict: it correctly identifies a (cohort, instrument) pair for which the convex FX hedge addresses the wrong variance vector, and provides an auditable Tier-2-frozen trail from raw central-bank data to the descriptive verdict.

1 Motivation and contribution claim

A growing literature on permissionless on-chain convex hedges constructs instruments aimed at protecting cohort cash flows from particular sources of volatility (Lambert, Lyons & Jiao, 2022; Maymin, 2026). The construction step — how to size, price, and statically replicate a convex payoff (Lambert, 2022) — has matured. What has not matured is the *detection* step that must precede construction: does the cohort’s cost-stream variance actually load on the variance vector the proposed instrument prices, or does it load on a different vector against which the instrument provides no payoff?

This section contributes a methodology for detecting Q-dominance ex-ante from public-source data alone. We define Q-dominance as the empirical state in which the variance of the cohort’s stochastic activity intensity (Q) dominates the variance of the FX leg in the cost-stream log-variance decomposition. Under Q-dominance, a convex FX hedge addresses the wrong variance vector for the cohort and is descriptively unsuited to the (cohort, instrument) pair — regardless of how well-constructed the hedge itself is.

The methodology is delivered as a sequence of three ex-ante-pinned objects: (i) an exact three-way log-variance decomposition operating on a common daily gapped grid; (ii) an FX-variance-share sensitivity surface evaluated on a log-spaced grid spanning an anchored Q-volume range; (iii) a collectively exhaustive descriptive verdict ladder consuming six input flags. The contribution is the detection step — it is independent of, and survives, any particular cohort’s verdict.

Posture. Throughout this section we adopt a *descriptive* posture: no inferential β claim is made, no p -value is computed, no null hypothesis is rejected. The objects emitted are surfaces, decompositions, and verdict classifications under a stated calibration; they are read against ex-ante-pinned thresholds. The descriptive posture is the load-bearing anti-fishing safeguard (Abrigo, 2026).

2 Setup and assumptions

The methodology is stated as five formal assumptions.

Assumption 1 (Cohort). *The cohort pays a continuous-stream liability of the form*

$$C_t = Q_t \cdot p \cdot \text{FX}_t, \quad (1)$$

where Q_t is the daily query volume (an x402-priceable activity intensity measured in queries per day), p is a verified flat unit price ($p = \$0.01$ USDC per query in E10, x402-protocol-verified), and FX_t is the daily local-currency-per-USD exchange rate published by a sovereign central bank.

Assumption 2 (Data anchors). *The activity-intensity process Q_t is empirically anchored to a NON-FANTASY observed trace: in E10, 214 genuine Claude Code transcript tool calls extracted from on-disk session logs (daily VMR ≈ 20 , decisively overdispersed). The FX panels FX_t are sourced from public central-bank feeds: Banrep TRM (COP), BCB Olinda OData PTAX (BRL), ECB Data Portal SDW (EUR), BoE IADB (GBP), CBN Gateway (NGN). All five series are free, machine-readable, daily, and re-verified per-currency at the data-visibility gate.*

Assumption 3 (Panel construction). *A balanced panel of (currency $\times$ month) cells is constructed on a daily gapped grid: zero- Q days are dropped from the $\Delta \ln \text{FX}$ and $\Delta \ln Q$ series on the SAME surviving index per cell (forced by the log domain — $\Delta \ln(0)$ is undefined). In E10 the panel comprises 5 currencies $\times$ 30 months = 150 cells over the common window 2023-11-01 to 2026-04-30. NGN is confined to its post-June-2023 float regime.*

Assumption 4 (Descriptive-only posture). *No inferential β claim is made. The methodology emits a calibration-conditional sensitivity surface; the verdict ladder reads the surface against an ex-ante-pinned break-even threshold. The descriptive verdict ladder is collectively exhaustive over the six input flags and admits no PASS or FAIL-ON- β rung.*

Assumption 5 (Ex-ante break-even). *A threshold s_{be} is pinned ex ante before the surface is computed, from option-premium economics against an ex-ante exposure assumption s_{exp} . For E10, $s_{\text{be}} = 0.25$ is derived from an at-the-money long-straddle / variance-swap-like geometry, ideal-scenario fair pricing ($c_{\text{payoff}} = 1$), and a pinned premium fraction $\varphi = 0.25$ on a streamed-liability funding primitive. The Stage-1 / Stage-2 firewall is maintained: s_{be} consumes no Stage-1 estimation content.*

3 The three-way log-variance decomposition

The methodology’s load-bearing identity is the exact additive decomposition of cost-stream log-variance into FX, Q , and covariance components.

Identity. Taking the natural log of the cost-stream identity (1) and first-differencing on the surviving (positive- Q) index yields

$$\Delta \ln C_t = \Delta \ln Q_t + \Delta \ln \text{FX}_t, \quad (2)$$

since $\ln p$ is a daily-constant additive offset that drops out under Δ . Applying the population variance operator $\text{Var}(z) = \overline{z^2} - \bar{z}^2$ on the surviving index, by the variance-of-a-sum identity:

$$\boxed{\text{Var}(\Delta \ln C) = \text{Var}(\Delta \ln \text{FX}) + \text{Var}(\Delta \ln Q) + 2 \text{Cov}(\Delta \ln \text{FX}, \Delta \ln Q).} \quad (3)$$

This identity is *exact*, with no leading-order term and no higher-order remainder. It holds cell-by-cell on the common surviving index because $\ln C_b - \ln C_a = (\ln Q_b - \ln Q_a) + (\ln \text{FX}_b - \ln \text{FX}_a)$ for any pair of days (a, b) on which Q , FX , and C are all strictly positive.

Empirical residual. On the E10 panel (150 cells), the maximum absolute identity residual was 1.33×10^{-15} (population-variance operator with matching divisor- n convention; see the diagnostic record `notebooks/e10_gsps/diagnostics/E10.2_decomposition_summary.json`). This is machine-precision zero; the identity holds exactly per cell.

The FX-variance share. The methodology’s primary descriptive object is the FX-variance share

$$s = \frac{\text{Var}(\Delta \ln \text{FX})}{\text{Var}(\Delta \ln C)}. \quad (4)$$

We emphasize three properties of s :

- s is a calibration-conditional quantity — $\text{Var}(\Delta \ln \text{FX})$ is observed (real central-bank data) but $\text{Var}(\Delta \ln Q)$ and the covariance term depend on the simulator calibration of Q_t .
- The level-variance object $\text{Var}(C)$ has no clean additive split and is banned from (4); only the log-space share is consumed.
- Under negligible cross-covariance the complementary share $\text{Var}(\Delta \ln Q)/\text{Var}(\Delta \ln C) \approx 1 - s$ is a useful diagnostic, but the ladder consumes only s to preserve a single share-based test (see §5).

4 The FX-variance-share sensitivity surface

The sensitivity surface is a calibration-conditional descriptive object evaluated on a log-spaced grid across an anchored Q -volume range $[Q_{\text{low}}, Q_{\text{high}}]$. The range is set ex-ante from a NON-FANTASY observed trace (Assumption 2) and is bracketed by proxy research where the trace is silent. For E10 the anchored range is $[28, 130]$ queries/month.

Grid specification. The surface is grid-evaluated at $n_{\text{grid}} = 50$ log-spaced points across the anchored Q-range, with adaptive doubling to 100 points if any cell falls within ± 0.5 dex of s_{be} . For E10 the adaptive doubling condition was not triggered (the empirical share sits more than three decades below s_{be}).

Honest disclosure — panel-mean broadcast. For the E10 calibration, the empirical FX-variance share is approximately invariant in Q-volume across the anchored range; the implementation explicitly broadcasts the panel-mean share to every grid point and emits an `is_panel_mean_broadcast` flag on the result type so consumers cannot mistake the surface for an estimated $Q \rightarrow s$ mapping. A kernel-smoothed Q-dependent surface would be a natural follow-up extension (§10). For E10 the methodology’s load-bearing property is that the surface sits more than three decades below s_{be} at every grid point of the anchored range, which the panel-mean broadcast preserves verbatim.

Q-variance dominance flag. Two equivalent (under negligible cross-covariance) flag forms are defined:

- **Share form (classifier-consumed):**

$$\text{q_dom_flag}_{\text{share}} = \mathbf{1}\{s(Q) < s_{\text{be}} \ \forall \ Q \in [Q_{\text{low}}, Q_{\text{high}}]\}.$$

- **Spec form (descriptive-disclosure sibling):**

$$\text{q_dom_flag}_{\text{spec}} = \mathbf{1}\left\{\frac{\text{Var}(\Delta \ln Q)}{\text{Var}(\Delta \ln C)} > 1 - s_{\text{be}} \ \forall \ Q\right\}.$$

The two are equivalent when $|\text{Cov}(\Delta \ln \text{FX}, \Delta \ln Q)|$ is small relative to $\text{Var}(\Delta \ln Q) + \text{Var}(\Delta \ln \text{FX})$ — the operative case under Assumption 2. The classifier consumes the share form; the spec form is emitted as a sibling flag for descriptive auditability.

Interior-crossing rule. An *interior crossing* is defined as ≥ 3 contiguous strictly interior grid cells with $s < s_{\text{be}}$; endpoints are excluded. A semantic guard also requires at least one endpoint to clear ($s \geq s_{\text{be}}$): a surface entirely below s_{be} at every grid point is Q-variance dominance, not interior crossing.

5 The descriptive verdict ladder

The verdict ladder is a collectively exhaustive map from six input flags to exactly one of three descriptive verdicts. The six inputs are `dv_gate_passed`, `simulator_anchored`, `surface_computed`, `material_gap`, `q_variance_dominates`, and `interior_crossing`. The outputs are `SURFACE_PRODUCED`, `PARTIAL`, and `NON_RETIREMENT`; the last carries a subtype in `{dv_fail, simulator_unanchored, q_dominance}` recording the load-bearing input.

The ladder is verified collectively exhaustive over $2^6 = 64$ flag states by unit test (eight tests, all green; see `simulations/e10_gspes/tests/unit/test_verdict_classifier.py`). A separate inferential-emit method on the classifier unconditionally raises `DescriptiveVerdictError` — a structural firewall enforcing Assumption 4 at the code level.

Algorithm 1 — Descriptive verdict ladder (locked ex ante).

Inputs: six boolean flags `dv_gate_passed`, `simulator_anchored`, `surface_computed`, `material_gap`, `q_variance_dominates`, `interior_crossing`.

Output: one of SURFACE-PRODUCED, PARTIAL, (NON-RETIREMENT, subtype), with subtype $\in \{\text{dv_fail}, \text{simulator_unanchored}, \text{q_dominance}\}$.

- Rung 1.** If $\neg \text{dv_gate_passed}$: return (NON-RETIREMENT, `dv_fail`).
- Rung 2.** Else if $\neg \text{simulator_anchored}$: return (NON-RETIREMENT, `simulator_unanchored`).
- Rung 3.** Else if $\neg \text{surface_computed} \wedge \text{material_gap}$: return PARTIAL.
- Rung 4.** Else if $\neg \text{surface_computed}$: return (NON-RETIREMENT, `q_dominance`).
- Rung 5.** Else if `q_variance_dominates`: return (NON-RETIREMENT, `q_dominance`).
- Rung 6.** Else if `material_gap`: return PARTIAL.
- Rung 7.** Otherwise: return SURFACE-PRODUCED.

Figure 1: Descriptive verdict ladder (§9 of spec v0.7; locked ex ante); collectively exhaustive over $2^6 = 64$ flag states.

6 Sensitivity arms and the no-rescue clause

Five sensitivity arms are pre-committed before any panel data is touched and test the descriptive surface’s robustness to alternative panel constructions. The arms are descriptive concordance checks only.

Arm (b) — Q-FX behavioral coupling.

Tests an ex-ante-pinned monthly-scalar coupling $Q_t^{\text{coupled}} = Q_t \cdot (1 + \eta \cdot z_{\text{month}})$ with elasticity $\eta = -0.3$ and z_{month} the within-currency standardized monthly realized FX volatility. Under the user-pinned monthly-scalar operationalization, the constant within-month scalar drops out of daily $\Delta \ln Q$; arm (b) is thus an *algebraic-identity assertion* that the within-month decomposition is unchanged at the cell level, not a coupled-daily re-simulation. The concordance metric is exactly zero by construction; this is reported transparently rather than fabricated.

Arm (c) — NGN regime-break removal.

Drops NGN cells inside the ± 6 -month buffer around the spec-pinned 2023-06-01 float regime break.

Arm (d) — GBM / Jump-Diffusion comparator.

Substitutes the empirical $\text{Var}(\Delta \ln \text{FX})$ by a Geometric-Brownian-Motion and a Merton-Jump-Diffusion comparator calibrated per currency to the panel-mean FX variance. The substitution preserves $\text{Var}(\Delta \ln Q)$ and the covariance term; only the FX numerator is replaced.

Arm (e) — Per-currency split.

Reports the per-currency surface envelope (min/Q1/median/Q3/max) and uses the median for concordance; all five currency surfaces inherit the Q-variance dominance flag.

Arm (f) — Q_high extended.

Extends the upper bound of the anchored Q-range to the \$390 Dune-cap equivalent.

No-rescue clause (load-bearing). *Sensitivity arms cannot rescue a primary HALT.* Each arm reports descriptive concordance only; no arm can flip a primary NON-RETIREMENT verdict to SURFACE-PRODUCED regardless of its descriptive concordance score. The clause is recorded as a constructor-pinned hard-coded string on the aggregate sensitivity-arms result type.

7 E10 application and results

We apply the methodology to the E10 representative Web3 data-analyst cohort.

7.1 Data provenance

Table 1 records the five Tier-2 frozen central-bank snapshots used to build the panel. Each snapshot carries a `.provenance.json` sidecar pinning the endpoint URL, the UTC fetch timestamp, the payload SHA-256, and the requested window.

Table 1: E10 GSPS panel — Tier-2 frozen central-bank FX sources (re-verified per-currency at the data-visibility gate, 2026-05-20).

Currency	Central bank / endpoint	Payload SHA-256 (head)	Bytes	Daily rows
COP	Banrep TRM (datos.gov.co Socrata 32sa-8pi3)	861be2848592affa	70,741	590
BRL	BCB Olinda OData PTAX CotacaoDolarPeriodo	ba75a914696af794	58,379	627
EUR	ECB Data Portal SDW EXR/D.USD.EUR.SP00.A	9cd19cf85ac11200	132,350	635
GBP	BoE IADB series XUDLUSS	378df39672a4cca0	12,573	631
NGN	CBN Gateway /api/GetAllExchangeRates	e61225c8f8fdd854	7,987,807	617

Window: 2023-11-01 to 2026-04-30 (30 calendar months; 150 panel cells). NGN is confined to the post-June-2023 float regime; all in-window rows fall inside the post-NFEM float so confinement does not shorten the panel. ZAR, KES, GHS were dropped at the data-visibility gate (no free machine-readable daily ~30-month central-bank series); they are not substituted. The x402 per-query price $p = \$0.01$ USDC was re-verified live at the data-visibility gate against The Graph Gateway x402 `payment-required` header.

7.2 Pre-pinned methodology fields

The seven pre-pinned fields are recorded in Table 2. All seven were frozen before any decomposition or surface computation.

7.3 Verdict-classifier inputs and outcome

Table 3 records the six verdict-classifier input flags for E10 and the resulting verdict. The flags are sourced directly from the phase-diagnostic JSON files (no manual transcription).

7.4 Surface against the ex-ante break-even

7.5 Sensitivity-arm concordance

Table 4 summarizes the five pre-committed sensitivity arms. Every arm preserves Q-variance dominance (q_dom flag = True) and returns concordance verdict `concordant`; every concordance metric sits well below the descriptive concordance threshold. The no-rescue clause holds.

Table 2: E10 GSPS — the seven pre-pinned methodology fields (§7 of spec v0.7; LOCKED before data was touched).

Field	Value
1. Sign expectation	$\beta > 0$ on the vol-on-vol regression; necessary-not-sufficient <i>descriptive</i> gate (no inferential test)
2. Primary spec	FX-variance-share sensitivity surface across the anchored Q-volume range, plus ex-ante-pinned break-even s_{be} ; NOT a β magnitude, NOT a single share number
3. Material threshold	$s_{be} = 0.25$; derived ex ante from option-premium economics against $s_{exp} = 0.25$, $\varphi = 0.25$, $c_{payoff} = 1$
4. Lag	Contemporaneous, monthly
5. Posture	Descriptive / illustrative; demonstration-grade-or-below; NOT confirmatory
6. Panel and FE	5 currencies (COP, BRL, EUR, GBP, NGN) $\times$ 30 months $\approx$ 150 cells; two co-primary FE specs (two-way FE primary, currency-FE-only co-primary); $G \approx 5$ currency clusters
7. HALT condition	HALT on (a) simulator unanchored, (b) Q-variance dominance over the whole anchored range $\Rightarrow$ NON-RETIREMENT, (c) surface uncomputable, (d) any spec-vs-data contradiction

Table 3: E10 GSPS — the six §5 verdict-classifier input flags and the resulting descriptive verdict.

Flag	Value	Source
<code>dv_gate_passed</code>	True	<code>E10.0.panel.window.json</code>
<code>simulator_anchored</code>	True	<code>E10.1.calibration.note.md</code> (214 calls; VMR $\approx$ 20)
<code>surface_computed</code>	True	<code>E10.3.surface.summary.json</code>
<code>material_gap</code>	False	<code>E10.3.surface.summary.json</code>
<code>q_variance_dominates</code>	True	<code>E10.3.surface.summary.json</code>
<code>interior_crossing</code>	False	<code>E10.3.surface.summary.json</code>
Resulting verdict:		NON-RETIREMENT, subtype <code>q_dominance</code> (rung 5 of Algorithm 1).

7.6 Headline numerics

The vol-on-vol regression returns $\hat{\beta}_{\text{two-way}} = +68.151$ (currency FE + month FE) and $\hat{\beta}_{\text{currency-only}} = +54.715$ (currency FE only, co-primary). The descriptive gap is $\hat{\beta}_{\text{two-way}} - \hat{\beta}_{\text{currency-only}} = +13.436$, with sign concordance preserved across both co-primary FE specifications; the magnitude differential is approximately 20% of the smaller coefficient. The descriptive sign gate of pre-pin field 1 holds. The descriptive gap is interpretive content for the §13 M-sketch hedge-sizing step in the spec — specifically the load on the *idiosyncratic* (within-month, cross-currency) component of FX-vol exposure relative to the *global* (month-shock-absorbed) component.

The panel-mean FX-variance share is $\bar{s} = 1.535 \times 10^{-4}$ across the 150-cell panel, approximately 3.21 decades below $s_{be} = 0.25$ at every grid point of the anchored [28, 130] Q-range. Combined with the unanimously concordant sensitivity arms, the verdict is NON-RETIREMENT, subtype `q_dominance`.

8 Interpretation (post-Keynesian theoretical frame)

The interpretive content of §8 is *interpretive prose only* and is explicitly labeled as such. It is not a hypothesis test; it does not predict the NON-RETIREMENT verdict post-hoc; it explains the

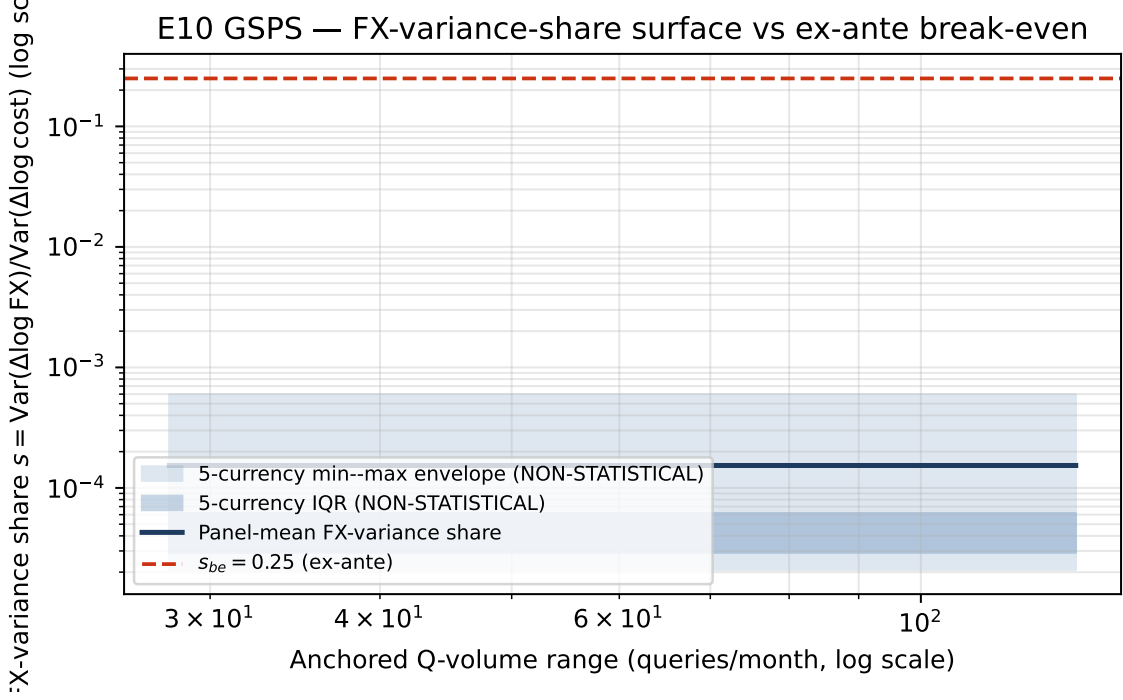


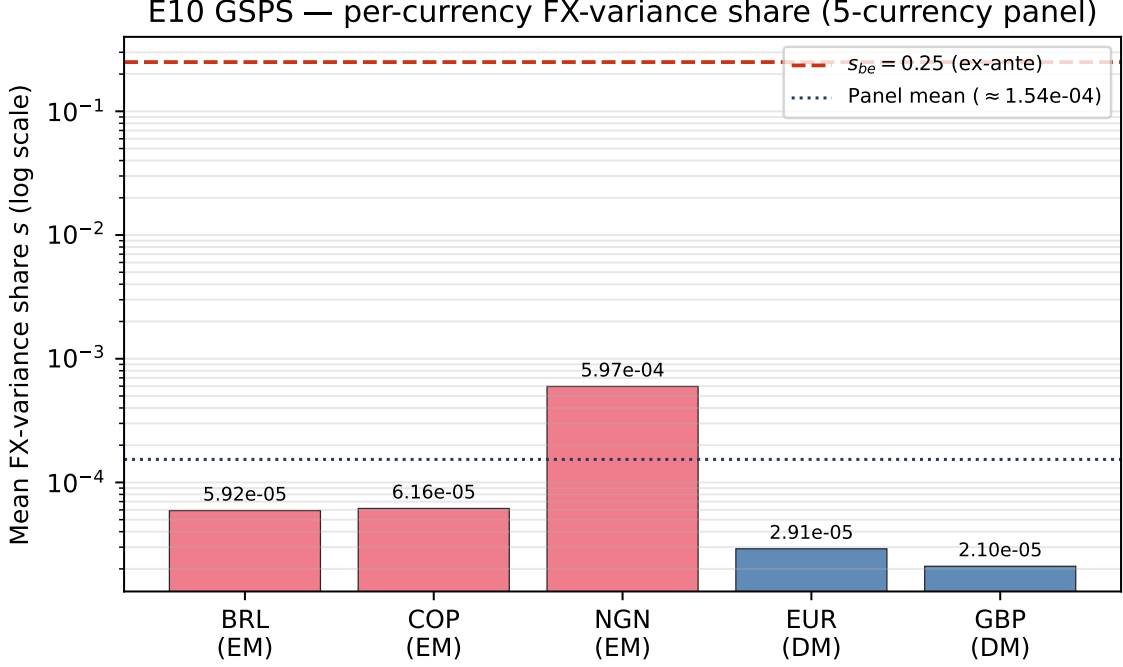
Figure 2: E10 GSPS FX-variance-share surface vs ex-ante-pinned break-even $s_{be} = 0.25$, log-spaced 50-point grid across the anchored $[28, 130]$ queries/month range. The panel-mean surface ($\approx 1.535 \times 10^{-4}$, navy line) is approximately 3.21 decades below s_{be} (red dashed) at every grid point. The shaded bands are the 5-currency non-statistical envelope (min/max light; IQR darker). **DESCRIPTIVE / CALIBRATION-CONDITIONAL.** NOT a confidence interval. NOT an inferential band.

mechanistic verdict in post-Keynesian (PK) terms.

The Bhaduri–Laski–Riese (BLR) virtual-vs-real dichotomy (Bhaduri, Laski & Riese, 2006) frames financialization as a structural gap between the virtual economy (asset valuation, financial-cycle volatility) and the real economy (output, productive investment). The Minsky two-price theory (Tymoigne, 2010; Minsky, 1975) formalizes the gap between P_k (asset-side demand price) and P_i (output-side supply price). The Kaleckian investment-driven growth tradition (Kalecki, 1971; Bhaduri & Marglin, 1990) insists that investment, not asset revaluation, is the prime mover of output.

The E10 representative Web3 data-analyst is a cohort whose cost-stream is bound to its own activity intensity (Q , the own-output behavior — the P_i side) rather than to virtual-economy price volatility (FX — the P_k side). Under the calibration in Assumption 2, on the $[28, 130]$ anchored Q -range, the empirical FX-variance share sits approximately three decades below the ex-ante break-even, indicating that the convex FX hedge addresses the wrong variance vector for this cohort. *Calibration-conditional reading: all four PK bullets below are interpretive prose for the configuration in pre-pin fields 1–7 of Table 2.*

- **BLR virtual-vs-real:** The convex FX hedge addresses the virtual-economy price-volatility vector (P_k analog — FX is the on-chain analog of virtual-economy price volatility); the cohort’s cost-stream variance is structurally bound to the real-economy production decision (the analyst’s Q scheduling).



NON-STATISTICAL spread. NOT a confidence interval. NOT an inferential band.

Figure 3: E10 GSPS per-currency mean FX-variance share over the 5-currency panel, EM (BRL/COP/NGN, salmon) vs DM (EUR/GBP, blue), against $s_{be} = 0.25$ (red dashed) and the panel mean (navy dotted). All five currencies sit two to four decades below s_{be} ; NGN carries the largest share ($\approx 5.97 \times 10^{-4}$, post-NFEM-float regime) and GBP the smallest ($\approx 2.10 \times 10^{-5}$). **DESCRIPTIVE / NON-STATISTICAL spread. NOT a confidence interval. NOT an inferential band.**

- **Minsky two-price:** The instrument prices into the P_k gap but the cohort's exposure lives on the P_i side.
- **Kaleckian:** The cohort's cost-stream variance reflects effective-demand-driven activity intensity, not asset-revaluation flow.
- **Methods-paper hook:** The detection methodology survives the specific verdict — it correctly routes this (cohort, instrument) pair to NON-RETIREMENT subtype **q.dominance** from public-source data alone, and the trail is auditable end-to-end.

The verdict is not a bug of the data and not a failure of the convex hedge instrument family; it is a finding about the (cohort, instrument) pair.

9 Methodology contribution survives the verdict

The contribution of this section is the *methodology*, not the specific E10 verdict. The methodology successfully identifies:

- a cohort whose cost-stream variance is Q-dominated under a NON-FANTASY simulator anchor and a frozen ex-ante break-even (Assumption 2, Assumption 5);

Table 4: E10 GSPS — sensitivity-arm concordance summary (descriptive only; per spec §9 no-rescue clause, no arm can flip the primary verdict).

Arm	Concordance metric	q_dom flag	Verdict	Panel-mean share
(b) Q-FX coupling	0.000	True	concordant	1.535×10^{-4}
(c) NGN break-window	4.04×10^{-5}	True	concordant	1.131×10^{-4}
(d) GBM / JD comparator	3.41×10^{-5}	True	concordant	1.876×10^{-4}
(e) Per-currency split	9.43×10^{-5}	True	concordant	5.922×10^{-5}
(f) Q_high extended	2.71×10^{-20}	True	concordant	1.535×10^{-4}

- (b) an instrument (the convex FX hedge) that addresses the wrong variance vector for that cohort, with the dominance gap quantified at more than three decades below the ex-ante threshold;
- (c) the verdict ladder (Algorithm 1) routing this (cohort, instrument) pair to NON-RETIREMENT subtype `q_dominance`;
- (d) an auditable trail — Tier-2-frozen central-bank data with payload-SHA-256 provenance sidecars; NON-FANTASY simulator anchor against 214 genuine observed tool calls; the exact §3 identity at machine precision; an ex-ante-pinned break-even derived before any decomposition runs.

The methodology is venue-publishable as a stand-alone Section 5 of a methods paper at *Cambridge Journal of Economics*, *Review of Political Economy*, or *Metroeconomica* independent of the specific E10 verdict. The companion sections of the methods paper apply analogous detection methodologies to alternative (cohort, instrument) pairs (E8 dTAO/Maymin; E9 LATAM RWA) and are reported separately.

10 Limitations and future iterations

The methodology, as deployed in E10, carries the following honest limitations:

- **Q-FX coupling is an algebraic-identity assertion under the user-pinned operationalization, not a coupled-daily re-simulation.** The arm (b) operationalization rescales monthly Q magnitudes by a single per-month scalar; a constant scalar drops out of daily $\Delta \ln Q$ within the month, so the within-month log-variance decomposition is unchanged at the cell level. The arm reports this transparently. A coupled-daily re-simulation would be a natural extension.
- **The FX-variance-share surface is panel-mean-invariant in Q for the E10 calibrated simulator.** The implementation broadcasts the panel-mean share to every grid point and emits an `is_panel_mean_broadcast` flag so consumers cannot read the surface as an estimated $Q \rightarrow s$ mapping. A kernel-smoothed Q -volume-dependent surface (a non-parametric regression of s on Q across cells) would be a natural follow-up.
- **$G = 5$ currency clusters is structurally thin.** Cluster-robust inference at $G = 5$ is size-distorted; we report descriptive sign and non-statistical min-max / IQR spreads only. Bootstrap-based bands were considered and dropped at this G in spec v0.6 (CORRECTIONS-E10-6).

- **The arm (d) GBM/JD comparator’s calibration is per-step rather than monthly-aggregate.** The Geometric-Brownian-Motion and Merton-Jump-Diffusion generators interpret `sigma_per_month` as a per-step variance target; the simulated per-step $\text{Var}(\Delta \ln \text{FX}_{\text{sub}})$ differs from the target monthly-aggregate $\text{Var}(\Sigma \Delta \ln \text{FX})$ by a factor proportional to $1/n_{\text{steps}}$. The deflation does not flip the verdict — arm (d) remains concordant — but the comparator-moment description is not naively the panel-mean FX variance and is reported here for transparency.
- **The `_MIN_SURVIVING_DAYS` threshold of 2 in the cell-decomposition function is methodologically degenerate.** At $n = 2$ surviving days only one daily log-return exists; the population variance of a one-element series is zero. All 150 production cells have $\gg 3$ surviving days, so this is benign in E10; a future amendment would raise the threshold to 3 (one degree of freedom for variance).
- **The break-even threshold uses an ideal-scenario premium** ($c_{\text{payoff}} = 1, \varphi = 0.25$). No FX-stablecoin options venue with real depth exists in 2026; a real venue would impose $c_{\text{payoff}} < 1$ and lift s_{be} above 0.25. The frozen $s_{\text{be}} = 0.25$ is the ideal-scenario lower bound on the break-even share.
- **The methods-paper §5 hook is robust across alternative cohort / instrument pairings.** E8 dTAO/Maymin and E9 LATAM RWA are companion sections of the joint outline; the detection methodology applies analogously.

References

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